

Preliminary Design Review

PSYCHE Capstone



*SES*410*

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Team Contributions

- Elysian Hudson - Team Leader
- Caleb Zick - Model Lead
- Daniel Joseph - Simulation Lead
- Jacob Coons - Computer Lead
- Spencer Emmons - Science Lead
- Syed Hamza Ayaz - Testing Lead



Agenda

- Science Question
- Background
- Motivation
- Science Traceability Matrix (STM)
- CONOPS
- System Block Diagram
- Requirement Flowdown
 - Mission-Level Requirements
 - Subsystem Definitions and Requirements
- Conclusion
- Next steps



Science Question



- ◆ By utilizing computer simulation of a hypothetical model of 16-Psyche's interior using differing initial conditions, determine the feasibility of directly mapping subsurface structures by imparting and recording seismic waves via an artificial impactor and seismograph.



Background - 16 Psyche



- 16 Psyche is a large asteroid within the main asteroid belt, presumed to be largely composed of metal & hypothesized to be the partial core of a planetesimal.
- Named after the Greek goddess of the soul.
- An orbiter mission to investigate the asteroid is currently underway, scheduled to arrive in 2029.
- Current NASA Psyche mission:
 - Instrument Suite: Multispectral Imager, Gamma-ray and neutron spectrometer, Magnetometer, Gravity science investigation
- Mission precedents: NASA's InSight, Deep Impact, LCROSS missions. JAXA's Hayabusa2 mission.
- Single-site and/or multi-site seismology used to develop 3-dimensional maps of subsurface structures of planetary bodies.

Background - Software Package



- What is SeisSol?
 - A software package for simulating wave propagation based on the arbitrary high-order accurate derivative discontinuous Galerkin method (ADER-DG)
- What does it do?
 - Utilizes meshes to approximate 3D model geometries
 - Specifies material elasticity to approximate geological subsurface properties
 - Produces reliable and sufficiently accurate synthetic seismograms and other seismological data sets
- Why does it work for us?
 - Can select source wave generation point(s) with whatever energy we decide
 - Can get detailed seismological data at any selected detection point (ground displacement and wave energy) in order to answer our science question.

SeisSol

Motivation



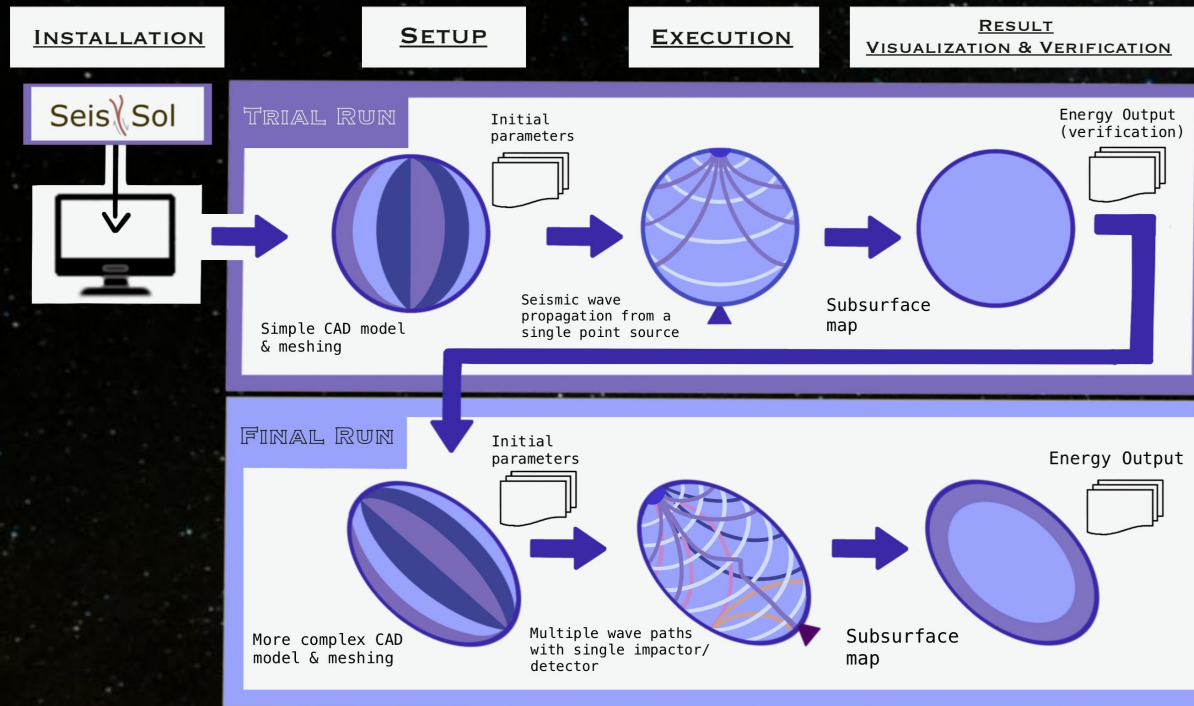
- Very little is known about the interior structures of small planetary bodies.
- Knowledge about 16 Psyche currently relies on remote sensing observations.
- Directly investigate 16 Psyche to help determine if it is the core of a planetesimal, using geophysics and seismology to explore the subsurface.
- Possibly differentiated/layered.
- Learn more about the planetary formation/evolution.
- Future implications for planetary defense and resource exploitation.

Motivation–Decadal Survey

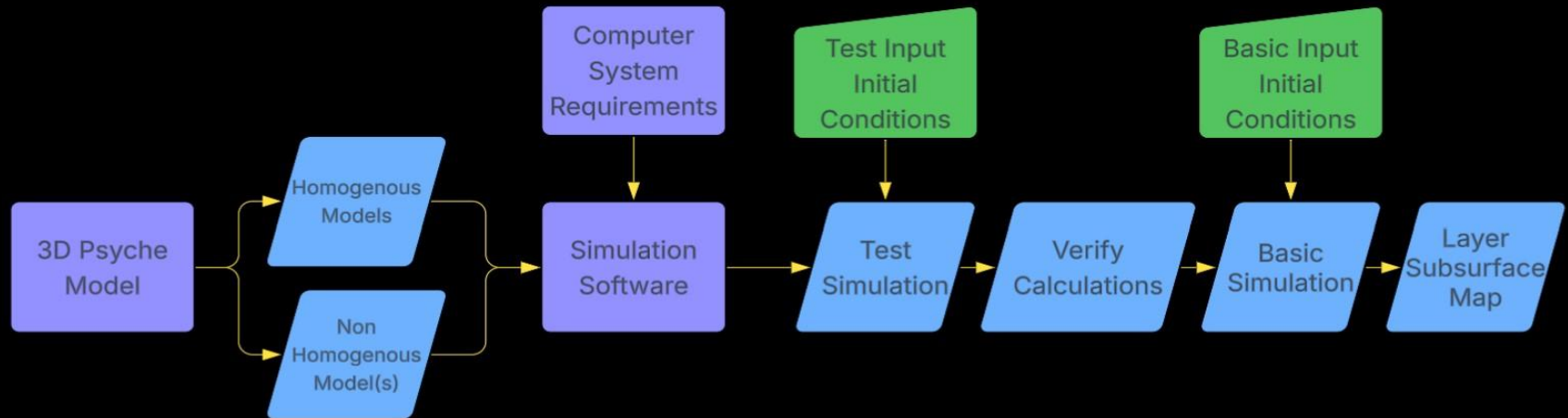


- 2023-2032 Decadal Survey Key Questions:
 - Q1 What were the initial conditions in the solar system, what processes led to the production of planetary building blocks, and what was the nature and evolution of these materials?
 - 16 Psyche is a presumed core, and therefore possibly a remnant planetary building block. Learning about its subsurface structure can tell us about the processes that lead to its formation, including processes such as differentiation.
 - Q4 How have the population of solar system bodies changed through time, how has bombardment varied across the solar system, and how have collisions affected the evolution of planetary bodies?
 - 16 Psyche may have experienced a hit-and-run event(s) during formation, and the subsurface structures can tell us about the impact history of the asteroid.
 - Q5 How did the interiors of solid bodies evolve, how is this evolution recorded in a body's physical and chemical properties, and how are solid surfaces shaped by internal and external processes?
 - 16 Psyche can be directly investigated to learn about the internal evolution of other solid bodies in our Solar System. Its internal structure will have recorded the processes responsible for its shaping.

Psyche CONOPS



System Block Diagram



Science Traceability Matrix (STM)



Objective	Measurement	Measurement Requirements	Technologies	Technological Requirements	Data Product
Simulate wave propagation through a model of 16 Psyche	Simulated seismic waves that fully travel through the interior of the model	Calculated minimum kinetic energy of the impact is 4 MJ	-Software -CPU -Memory	<u>Software</u> -SeisSol Simulation Software and dependencies (see software subsystem slide)	Seismic wave energies at selected measurement point
	Detect simulated signal at selected point on the model of 16 Psyche	Minimum measurable ground acceleration spectral density of: $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$		<u>CPU</u> -Single chip vs cluster vs supercomputer setup (TBD) -Performance: minimum clock speed, # of cores, # of threads (TBD)	Surface displacement at selected measurement point
	3D Model of 16 Psyche's interior	Minimum accuracy of bilayer, non-homogeneous, order of magnitude density estimate of 16 Psyche		<u>Memory</u> -Minimum amount of RAM (TBD) -Minimum DDR Performance (TBD)	Animated visualization of seismic waves traversing the model

Requirements Levels

L1: Mission and Measurement Requirements

L2: Subsystem Requirements



Requirements

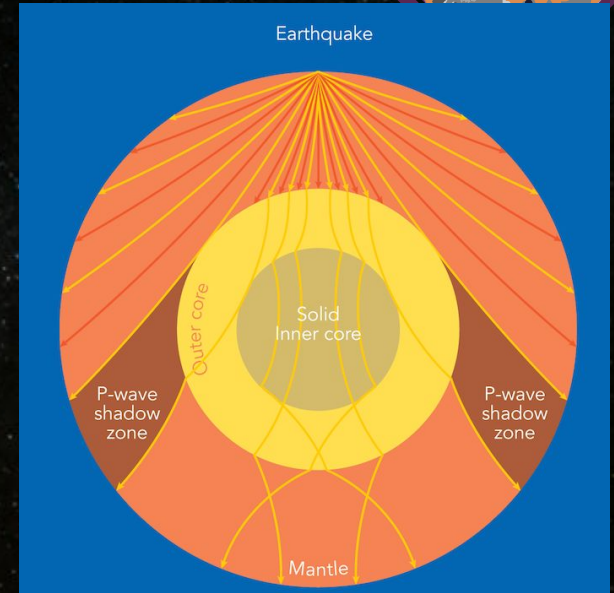


Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
L1.1	Propagation of seismic waves - The simulation shall accurately model the propagation of seismic waves through a 3d body.	Test, Analysis, Demonstration	Science Question
L1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	Test, Demonstration	Science Question
L1.3	Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments and simulated asteroid.	Inspection	Science Question

L1 Requirement – 1.1

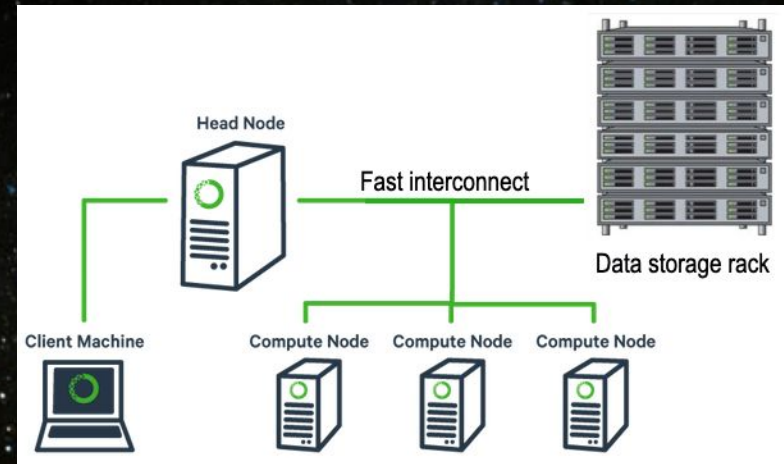
- Propagation of seismic waves - The simulation shall to accurately model the propagation of seismic waves through a 3d body
- Shall be tested using a simplified model and the results compared to manually calculated results
- Takes into account:
 - Wave attenuation
 - Reflections/refractions at boundaries



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.

L1 Requirement – 1.2

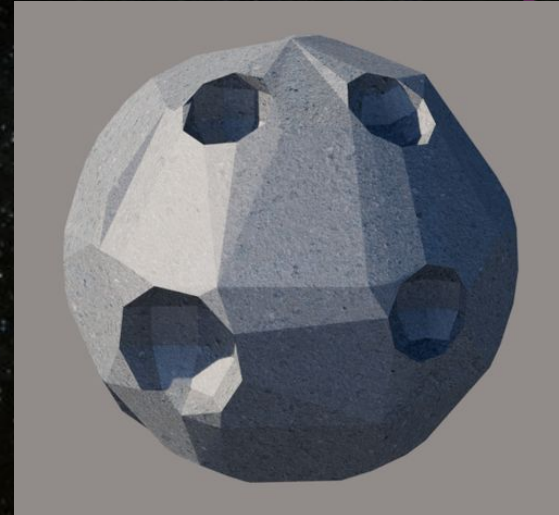
- Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)
- Considerations:
 - Hardware (accessibility, performance, compatibility with software)
 - Software (CPU architecture, package management, Operating System, file I/O)



Corredor, John & Rodriguez, Camilo. (2021). Beowulf Cluster to support disciplinary education and research in Computer Science.
10.13140/RG.2.2.36786.12481.

L1 Requirement – 1.3

- Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments and simulated asteroid.
- Includes:
 - Varying Model Characteristics:
 - Density
 - Porosity
 - Layering
 - Varying Inputs:
 - Initial Energy
 - Locations of “Impactor” and “Detector”
 - “Impact” angle (initial direction of waves)



Romanenko, K. (2019, June 17). *asteroid-5*.
GrabCAD.

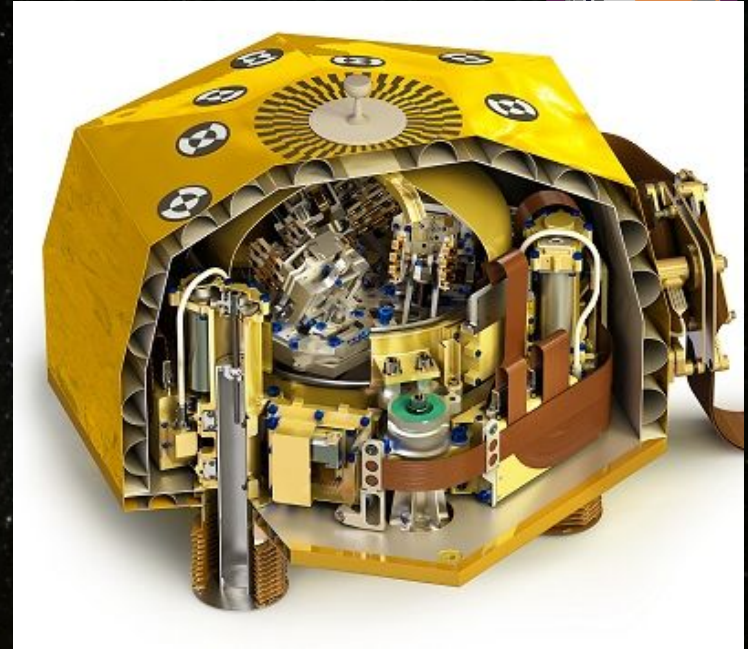
Subsystem Requirements - 1.1



ID	Requirement	Verification	Parent	Rationale
L2-WAVE-1	Basic tests of the simulation code shall use homogeneous spheres of various densities approximating 16 Psyche - a radius of 115 km	Test, Analysis	L1.1	Confirms the accuracy of the simulation code
L2-WAVE-2	The readings at the simulated detector shall be within the sensitivity limits of the SEIS instrument, with a ground displacement of 1 Å and a ground acceleration spectral density of $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$	Demonstration, Analysis	L1.1	Confirms the waves could be detected by a real-world seismometer

Science Subsystem

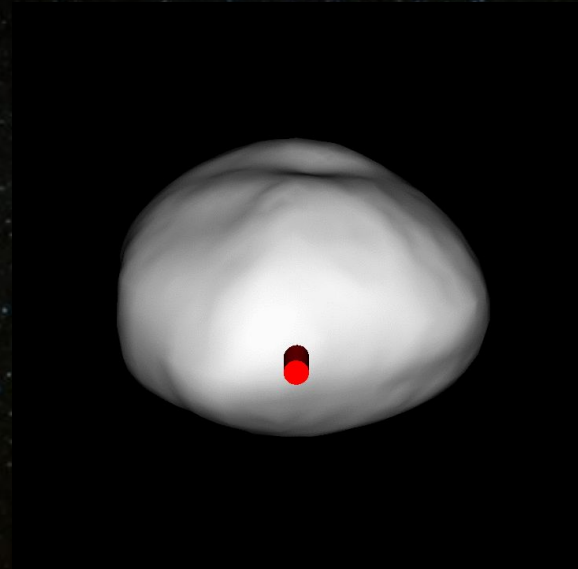
- Accuracy - The simulation code's accuracy will be confirmed by comparison to manually calculated results in simple examples
- Measurability - The simulated seismic waves at the location of the detector must be above the noise floor of real-world detectors
 - SEIS - The SEIS (Seismic Experiment for Internal Structure) is a seismometer on Mars included with the InSight Mission - due to its proven use on a non-Earth object it will be our real-world reference point
 - The SEIS is sensitive to a ground displacement of $1 \text{ \AA}$ and ground acceleration spectral density of $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$



Interactive VBB pendulum. L'instrument seis - SEIS / Mars InSight. (2018, February 26).

Model Subsystem

- To begin any simulation, SeisSol needs to internally construct a model, which can be accomplished via two methods:
 - Importing a CAD model, specifically in the .stl file format
 - Mathematically defining a basic shape in a YAML parameter file
- By using more basic shapes, we can verify the accuracy of the simulation by double-checking its results.
- Then, we can move up to more advanced shapes with confidence.



Shepard et al. (2021). Asteroid 16 psyche: Shape, features, and global map. The Planetary Science Journal, 2(4), 125.

Subsystem Requirements - 1.2



ID	Requirement	Verification	Parent	Rationale
L2-COMP-1	The OS and other dependency software shall support the successful operation of SeisSol	Test	L1.2	Prevents compilation and runtime errors
L2-COMP-2	The hardware models and setups shall meet the minimum performance required to successfully complete a run of SeisSol lasting no longer than 24 hours	Demonstration	L1.2	Allows for multiple runs of various 16 Psyche Models

Computer Subsystem - Software



- SeisSol - Our primary seismic wave simulation software.
- Operating System - SeisSol was developed on a unix-based OS. For maximum compatibility, we should run it on the same OS.
- Package Management - SeisSol utilizes Spack Open Source Package Manager for installation of the program, as well as all of its dependencies.
- CPU Architecture - Host architecture and corresponding command is needed for compiler tuning variables and utilization of proper code generators upon build phase, allowing for optimal build performance (TBD by hardware specs)

Computer Subsystem - Hardware



- CPU - Performance depends on multiple factors:
 - **Individual CPU performance:** Clock speed, number of cores, instructions per clock (IPC), number of threads for multitasking, etc (TBD)
 - **Number of CPUs in Setup:** One CPU vs a CPU cluster system vs ASU's Sol supercomputer. (TBD)
- Memory - Random Access Memory (RAM). Performance reliant on:
 - **Number of Gigabytes** - There is a minimum amount of RAM needed to meet our runtime demands (TBD)
 - **DDR (Double Data Rate)** - DDR Generation may matter more with less RAM, and less with more RAM (TBD)

Subsystem Requirements - 1.3



ID	Requirement	Verification	Parent	Rationale
L2-SIM-1	The simulation code shall be able to support varying numbers and positions of detectors	Inspection	L1.3	Shows effect of varying detectors on data
L2-SIM-2	The simulation code shall be able to support varying numbers, positions, energies, and impact angles of impactors	Inspection	L1.3	Shows effect of varying impactors on data
L2-SIM-3	The simulation code shall be able to simulate varying density, layering, impact, energy efficiency, and porosity of 16 Psyche models	Inspection	L1.3	Shows effect of varying conditions of 16 Psyche on data

Simulation Subsystem

- Custom CAD importing and meshing
- Simulate point source impact
- Allow for control over initial conditions (material density, impact source point, wave velocities)
- Visualize wave pathing through 3D Psyche model
- Output total energy of simulation



Testing Subsystem



- To confirm our results, we can compare the the energy from our test simulation with our calculations.
- The Seismic wave energies can be calculated by using the initial values by hand
- By using the values used for:
 - v_p = longitudinal wave velocity
 - v_s = transverse wave velocity
 - F = momentum of impactor
 - τ = time constant
 - ρ = density
- The total Seismic Wave Energy ($v_s + v_p$) equation can be separated into energies for the longitudinal and transverse waves

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Conclusions



- Our science question depends on a simulation of probable internal conditions and structural layers of 16-Psyche, to model our impactor energies and produce a measurable and significant detection with a seismograph.
- The subsystem requirements for this question necessitate the use of SeisSol's proficiencies and visualisation to model and test our assumptions of a theoretical subsurface and produce reasonable a map of 16-Psyche's interior.

Next Steps

- Follow through on PDR notes and begin preparation for CDR.
- Determine CPU and memory requirements.
- Get into contact with SeisSol developers regarding minimum requirements/science goal and overall scope of the project.
- Reach out to Dr. Heiner Igel, a professor of seismology at LMU, regarding possible simulation software alternatives if necessary.
- Look into possibly gaining access to computers on ASU.



**Thank
You!
Questions?**



Backup Slides:



Requirements Must be SMART:

- Specific -
 - It must address only one aspect of the system design or performance
 - It must be expressed in terms of the need (what and how well), not the solution (how).
- Measurable -
 - Performance is expressed objectively and quantitatively
 - For example, an exact pointing requirement (in degrees) can be tested thus verified prior to launch.
- Achievable -
 - It must be technically achievable at costs considered affordable
 - Note: JWST early designs specified an aperture requirement eventually descope due to technical issues with deployment.
- Relevant -
 - It must be appropriate for the level being specified
 - For example, requirement on the solar cells should not be designated at the spacecraft level.
- Traceable -
 - Lower level requirements (children) must clearly flow from and support higher level requirements (parents).
 - Requirements without a parent are referred to as orphans, and need to be assessed for necessity of inclusion.

STM Contents

(Groppi)

- Teams developing NASA missions use an STM to summarize the science and top-level requirements.
- An STM has the following columns:
 - **Science Objectives:** The elements of your science question(s)
 - **Measurement Objectives:** What quantities do you need to measure to achieve the science objective:
 - **Science Measurement Requirements:** What do you need to measure and how well do you need to measure it? Expected measurement range and required measurement error.
 - **Instrument Functional Requirements:** What other requirements does the instrument that takes the data need to have to meet your needs?
 - **System Functional Requirements:** What other requirements does the whole experiment need to have to meet your requirements?
 - **Science Product:** What is the data your experiment will produce (the low-level data measured, not the answer you get after you analyze it).

Basic simulation

It consists of:

- 1 Impact point
- 1 detection point
- 2 -3 layers

An example of SeisSol simulation for 2 layers

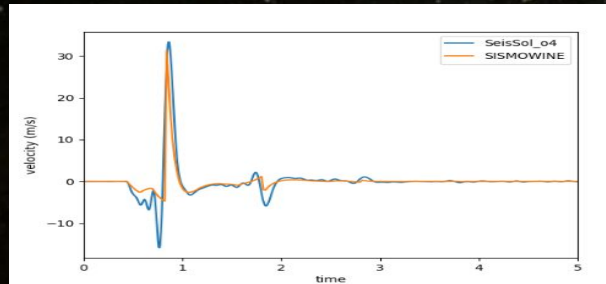
From central point (0,0,0)

Layer: from 0 to 1000 (z direction);

Half-space: 1000 to 3300 (z direction)

Impact source: (0, 0, 2000).

We can pick any impact and detection point by specifying the coordinates,

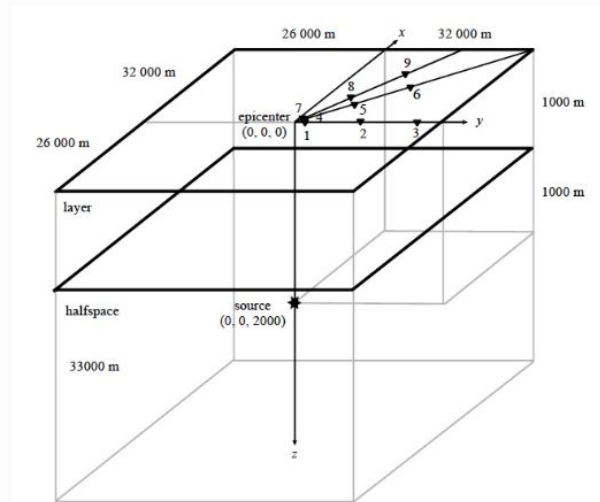


Example of a basic simulation of LOH1



	Vp (m/s)	Vs(m/s)	density	Qp	Qs
layer	4000	2000	2600	Inf	Inf
half-space	6000	3464	2700	Inf	Inf

Table: Material properties in LOH1.



Geometry of LOH1.

Requirements



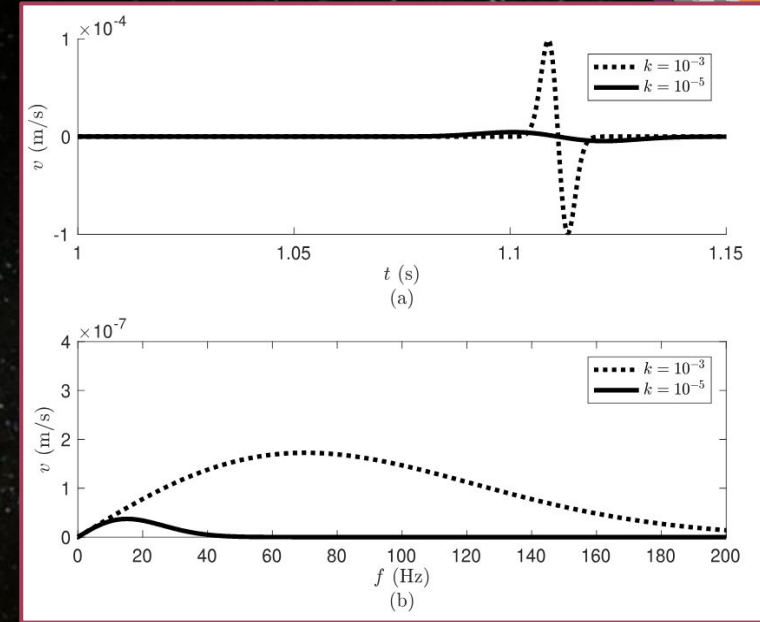
Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
MR1.1	Propagation of seismic waves - The simulation shall to accurately model the propagation of seismic waves through an artificial 3d body.	Output of test model matches manually calculated tests to 1 Order of Magnitude	Science Question
MR1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	The simulation completes 1 run in 24 hours or less	Science Question
MR1.3	Initial Conditions - The simulation code must be able to support varying initial conditions within the simulated instruments and simulated asteroid.	The simulation code can be run repeatedly with varying initial conditions	Science Question

Mission Requirement – 1.1



- Use previously acquired data to estimate frequency range, needed sampling rate, and timing accuracy.
- Use finite element models and seismic wave propagation simulations of Psyche's interior to refine previously estimated data.
- Use simulations to determine optimal seismometer placement.
- Measure changes in seismic signal characteristics as they propagate to map the subsurface interior.



Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.2



- Impact device shall generate enough energy to propagate seismic signal throughout the entirety of 16 Psyche.
- Requires multiple preliminary assumptions that are not yet known to be accurate to 16 Psyche.
- Simulation will be needed to better estimate 16 Psyche's wave velocities, seismic efficiency of impact, $k = E_s / E_k$, and other key values.

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

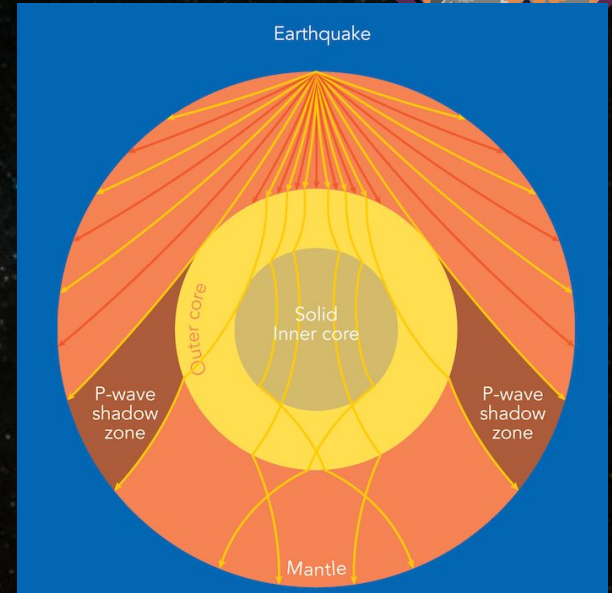
Where

- F = momentum of impactor
- v_p = longitudinal wave velocity
- v_s = transverse wave velocity
- τ = time constant
- ρ = density of target surface

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.3

- The simulation shall be able to account for varying numbers and positions of impactors and/or detectors.
- More impactors/detectors will improve precision of measurements.
- But larger numbers will also be less practical.



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.